Report - Effectiveness of isolation and contact tracing to control Bundibugyo virus (Ebola disease) outbreak in the Democratic Republic of the Congo

Joshua W. Lambert[1,2,*], Sebastian Funk[1,2,], and Carl A.B. Pearson[3]

[1]Department of Infectious Disease Epidemiology and Dynamics, London School of Hygiene & Tropical Medicine, London, United Kingdom
[2]Centre for Mathematical Modelling of Infectious Diseases, London School of Hygiene & Tropical Medicine, London, United Kingdom
[3]Department of Epidemiology, University of North Carolina, Chapel Hill, NC
[*]Corresponding author: Joshua W. Lambert, joshua.lambert@lshtm.ac.uk

May 28, 2026

Introduction

As of 24 May 2026 there were 906 cumulative suspected cases and 105 confirmed cases of Ebola virus disease caused by the Bundibugyo virus in the ongoing epidemic[1]. The outbreak is primarily in the eastern region of the Democratic Republic of the Congo (DRC), mostly in the Ituri Province, but with confirmed cases in North Kivu and South Kivu[1]. There are seven confirmed cases in Uganda, three imported from the DRC and others secondary cases among their contacts[2]. On 17 May 2026 the World Health Organization (WHO) declared the Bundibugyo virus outbreak a public health emergency of international concern (PHEIC)[3].

There is likely an underascertainment of the total cases involved in the Bundibugyo virus outbreak. A report from Imperial College London estimated the total outbreak size on 20 May 2026 was likely 400–900 cases, with a possibility of more than 1,000 cases, when the number of suspected cases reported was 516[4]. A second analysis of the true outbreak size used a joint modelling approach found the most likely total number of Ebola cases on 23 May 2026 to be 930 (90% CrI 483–2234)[5].

Genetic evidence points to a single spillover event with an outbreak start date around 11 April 2026[6].

Bundibugyo ebolavirus has no licensed vaccine or therapeutic, unlike Zaire ebolavirus, for which both a vaccine (Ervebo) and monoclonal antibody therapies (Inmazeb and Ebanga) have been approved[7]. With strong evidence of epidemic growth from increasing suspected cases[1] ($R_t > 1$), Public Health and Social Measures (PHSMs), a type of non-pharmaceutical intervention, are required to bring the outbreak under control. Isolation of suspected or confirmed (test positive) Ebola cases, and tracing of their contacts is a standard response strategy for Ebola outbreaks[8].

We assess options to control the ongoing Bundibugyo virus outbreak in the DRC using a stochastic individual-level branching-process model. We calibrate the model to current, and highly uncertain, surveillance data by conditioning simulated trajectories on the observed early outbreak size assuming a single spillover event. We then use the calibrated model to evaluate two responses: the proportion of contacts of known cases that are successfully traced (0%–100%), and the speed at which symptomatic cases are isolated after symptom onset (mean 1, 3 or 5 days). We simulate the introduction of various combinations of these components mid-outbreak, assuming PHSMs are activated at the declaration of the public health emergency, across plausible values of the basic reproduction number (R_0), to estimate the probability of bringing the outbreak under control within 3, 6 and 12 months of activating the response.

Methods

We use a stochastic individual-level branching process model of disease spread in the population. We assume a single index case initiates the outbreak and that transmission in the community is defined by a Negative Binomial offspring distribution with a mean equal to the basic reproduction number (R_0), and a dispersion parameter (k) to define the individual-level variability in transmission. At present there are no precise estimates for the R_0 for the current Bundibugyo outbreak. We set R_0 to 2.0 and 3.0, based on recent growth in suspected and confirmed cases [1], to explore the the sensitivity of our results to assumed Ebola transmissibility. There is little evidence to inform the individual-level heterogeneity in transmission for Bundibugyo virus. A systematic review of Ebola virus disease found that most estimates show moderate to high heterogeneity ($k < 1$) [9]. We use the estimate of Polonsky et al. [10] from the 2018–2020 Ebola epidemic in the DRC ($k = 0.27$).

Ebola cases are isolated after a delay from symptom onset. We model this delay from onset-to-isolation as a Gamma distribution. To evaluate how the speed of isolating cases with a positive test result impacts the ability to bring the outbreak under control, we run scenarios with a mean onset-to-isolation delay of 5 days, 3 days and 1 day (all assuming a standard deviation of 1 day). Cases can be isolated earlier than the delay after symptom onset if they are traced as a contact of a known case (i.e. their infector). When a contact is traced they are isolated as soon as their infector is isolated (or after their delay from symptom onset if earlier). Isolation is assumed to reduce all further transmission ($R_0^{\text{iso}} = 0$). We assess how contact tracing ascertainment influences the probability of bringing the outbreak under control, varying the percentage of contacts traced between 0% and 100% at 20% intervals.

To simulate the speed of transmission the model samples the generation time by first sampling each case's symptom onset time from the incubation period, then drawing the infection time of each secondary case from a skew-normal distribution centred on the infector's symptom onset and parameterised by the proportion of transmission that occurs before symptom onset [11]. We use an incubation period estimated from the 2014 Ebola outbreak in West Africa (Gamma distribution, mean = 10.3 days, SD = 8.2 days) [12]. This is similar to the reported incubation period of the Bundibugyo virus (2-21 days, [7]). We assume that no Bundibugyo virus transmission events occur prior to symptom onset (0% presymptomatic transmission), which is consistent with previous Ebola outbreaks [13]. Our epidemic model assumes a symptom-based isolation and contact tracing system. Therefore, asymptomatic cases are assumed to be missed by the intervention and continue to transmit in the community with the same transmissibility as symptomatic cases. We assume all Bundibugyo cases are symptomatic (0% asymptomatic). Again following evidence from past Ebola epidemics which found little evidence of asymptomatic or paucisymptomatic cases [14].

To simulate scenarios consistent with the ongoing Bundibugyo virus outbreak in the DRC, we conditioned each simulation (single index case) on having a cumulative case count in $[440, 2230]$ in at least one of weeks 5, 6 or 7 of the outbreak. Conditioning was implemented by rejection sampling: branching-process simulations were re-run with independent random seeds until each replicate produced a trajectory meeting the criterion, with up to 1,000 attempts permitted per accepted replicate.

We retained 200 accepted replicates per scenario. We assume that no PHSMs (i.e. isolation of cases or contact tracing) are active for the first 42 days of the outbreak (from the spillover infection). After day 42 we assume that individuals that are symptomatic or contact traced from a symptomatic individual are tested and isolated if they receive a positive test result to prevent further infections. We simulate the outbreak for a maximum of 2 years (730 days).

We assess the influence that not all untraced contacts are isolated. In a response that relies on confirmatory testing, imperfect test sensitivity means that a proportion of true Bundibugyo cases return a false-negative result and are not isolated. Alternatively, in a response that isolates suspected cases on clinical presentation alone, some Bundibugyo cases may go unrecognised and similarly fail to isolate. We model both situations using the same model parameter, *diagnostic sensitivity*, and vary it across 50%, 75% and 100%. Reduced sensitivity against Bundibugyo virus is plausible: there are no diagnostic assays specifically developed or validated for Bundibugyo, and the rapid antigen tests and most widely deployed PCR-based assays were developed and validated primarily against Zaire ebolavirus [15]. Symptomatic cases that are not isolated due to misdiagnosis continue to transmit in the community; cases isolated after symptom onset via contact-tracing follow-up are assume to always successfully isolate.

We define outbreak control as no new cases of Ebola virus disease from a certain date to the end of the outbreak. This is equivalent to extinction as we only have a single introduction of the virus into the human population with no subsequent spillover events. In our simulation setup we assume that if the cumulative number of infections exceeds 10,000 then that specific outbreak is considered uncontrollable. We test for outbreak control (extinction) at 3, 6 and 12 months from the activation of the PHSMs.

Results

The R_0 for transmission in the community impacts the ability to control this Bundibugyo Ebola epidemic (Figure 1). There is a remote chance ($< 5\%$) of ending the outbreak within 3 months from the activation of PHSMs, namely isolation of cases and contact tracing. Only in the scenario where $R_0 = 2$ and 100% of contacts for infectious individuals are traced does the outbreak have a highly unlikely ($> 15\%$) chance of ending (extinction) within 3 months with NPIs (Figure 1).

If R_0 is around or below 2 then contact tracing at least 80% of contacts shows a realistic to highly likely possibility of outbreak control within 6–12 months of PHSMs being active (Figure 1). If community R_0 is 3 then 100% of contacts need to be traced to have more than a remote chance of control, and even with 12 months of interventions, there is less than a highly likely chance of control (Figure 1).

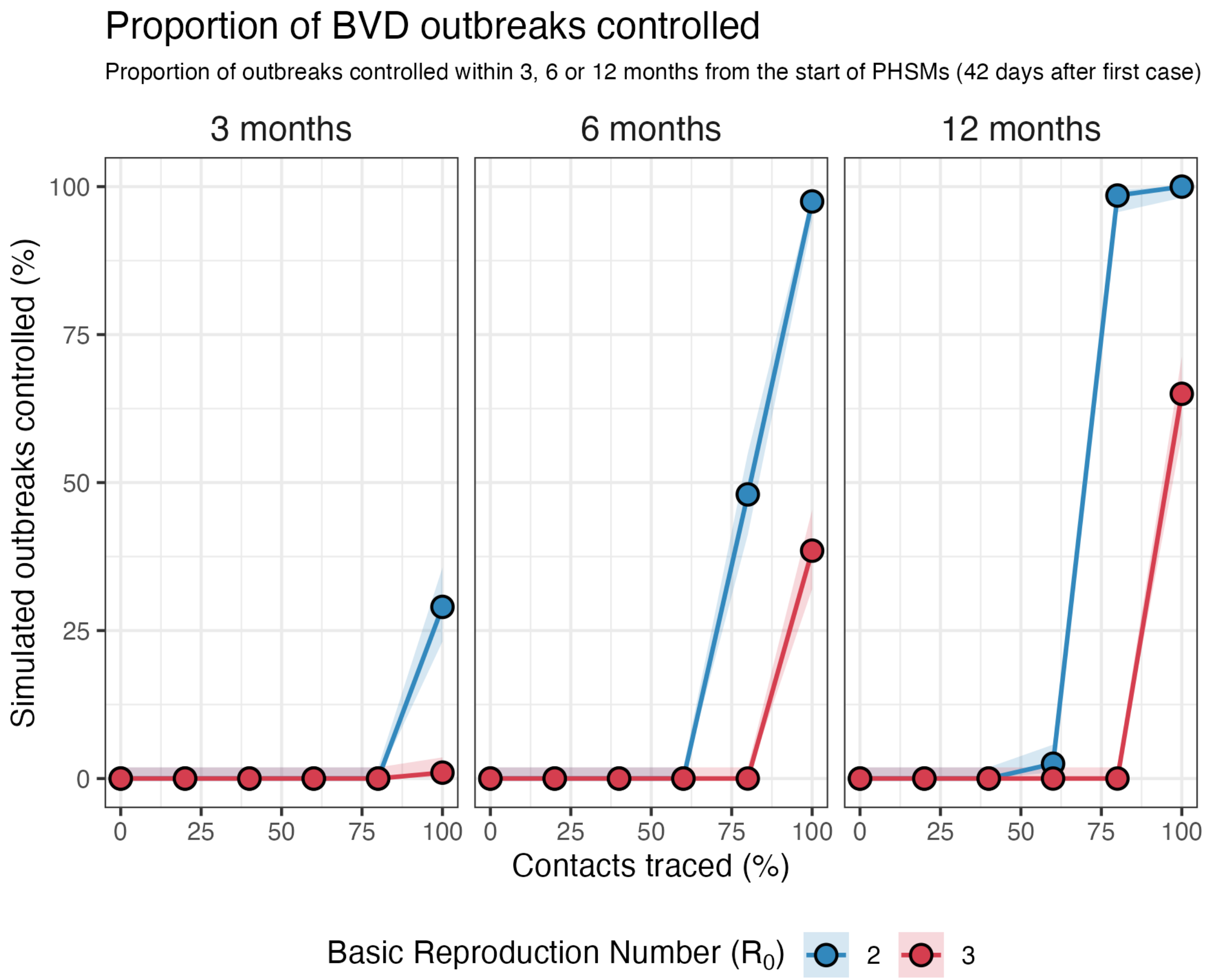

Figure 1: Proportion of simulated Bundibugyo virus outbreaks controlled (no new cases by the evaluation horizon) within 3, 6 and 12 months of PHSM activation, as a function of the percentage of infectious contacts traced (0%, 20%, 40%, 60%, 80%, 100%). Each point is the proportion of 200 conditioned simulations meeting the control criterion; shaded ribbons are 95% Wilson binomial confidence intervals around each proportion. PHSM activation is modelled as day 42 of the outbreak. The mean onset-to-isolation delay is fixed at 3 days (Gamma, SD = 1 day). Colours distinguish basic reproduction numbers R_0 = 2.0 and 3.0.

All results described above assume a mean time delay of 3 days from symptom onset to isolation. Assuming a longer mean delay to isolating cases of 5 days does not drastically alter the results of the 3-day delay, with probabilities of control marginally worse (for all scenarios not at 0% or 100% of outbreaks controlled) (Figure 2). If cases can be isolated on average 1 day after symptom onset and $R_0 \sim 2$, outbreak control is likely within a year if at least 20% of contacts are successfully traced. Outbreak control within 12 months is highly likely if contact tracing can ascertain $>$ 40% of contacts. When $R_0 > 2$ contact tracing needs to be comprehensive (close to 100%) for probable control within a year, and even then, the chance of control within 6 months is at best only a realistic possibility (Figure 2).

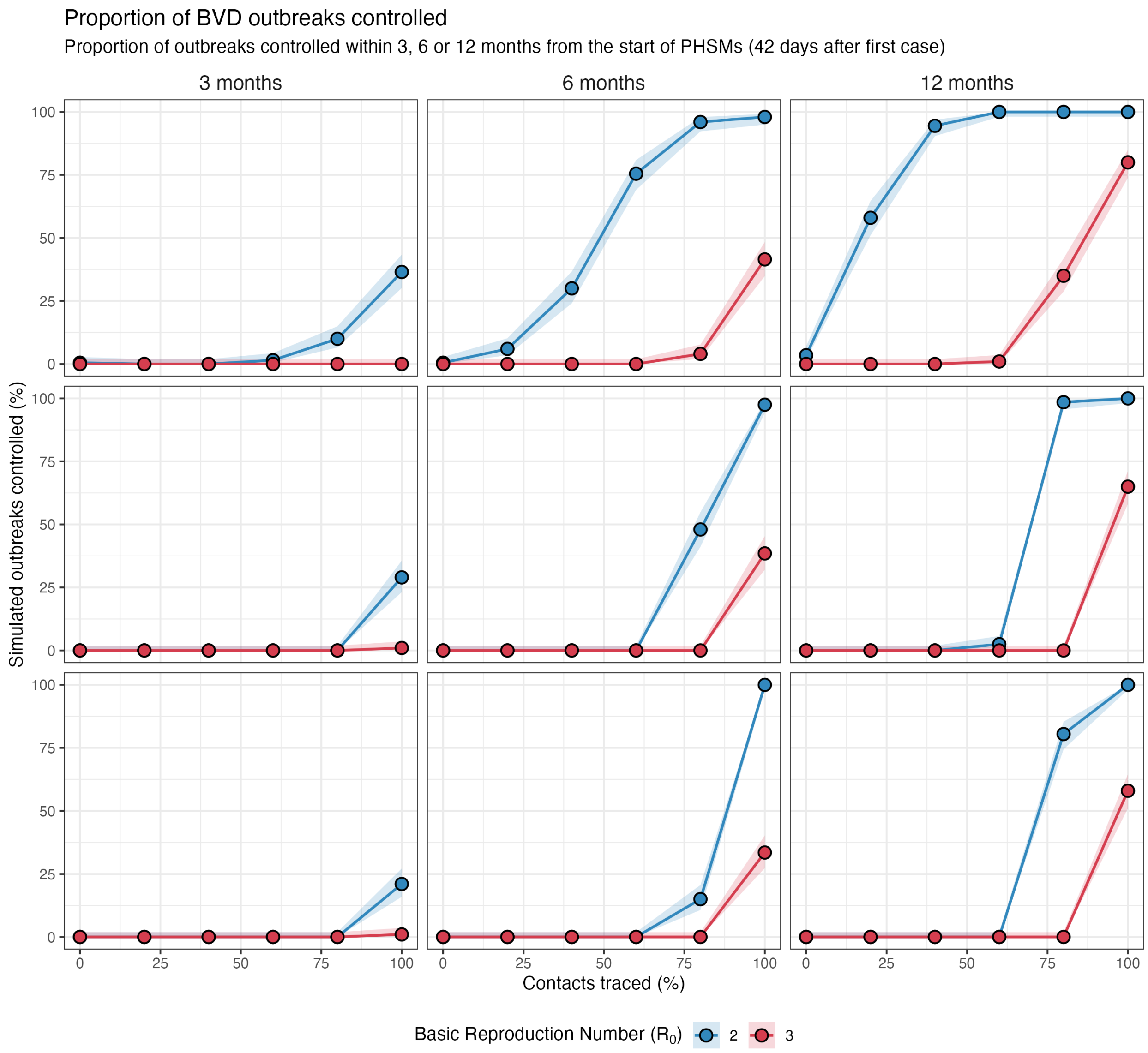

Figure 2: Proportion of simulated Bundibugyo virus outbreaks controlled (no new cases by the evaluation horizon) within 3, 6 and 12 months (columns) of PHSM activation, as a function of the percentage of infectious contacts traced. Rows show three mean onset-to-isolation delays (top to bottom: 1, 3 and 5 days; Gamma, SD = 1 day). Each point is the proportion of 200 conditioned simulations meeting the control criterion; shaded ribbons are 95% Wilson binomial confidence intervals around each proportion. PHSM activation is modelled as day 42 of the outbreak. Colours distinguish basic reproduction numbers R_0 = 2.0 and 3.0.

The sensitivity to correctly diagnose Ebola virus disease cases, either from a positive test or a suspected/probable case based on symptoms, can have a substantial impact on the ability for PHSMs to contain the outbreak (Figure 3). Notably, in a scenario that all community cases are correctly identified, the chance of outbreak control within 3 months, assuming a 1 day average delay from symptom onset to isolation, are greatly improved (Figure 3 topleft panel), and there is an almost certain chance of containment within 12 months at any contact tracing coverage if R_0 $\sim$2 (Figure 3 topright panel). Across all symptom onset to isolation delays and R_0 values modelled, when diagnostic sensitivity is 50% then the chance of outbreak control with 12 months is remote (Figure 3).

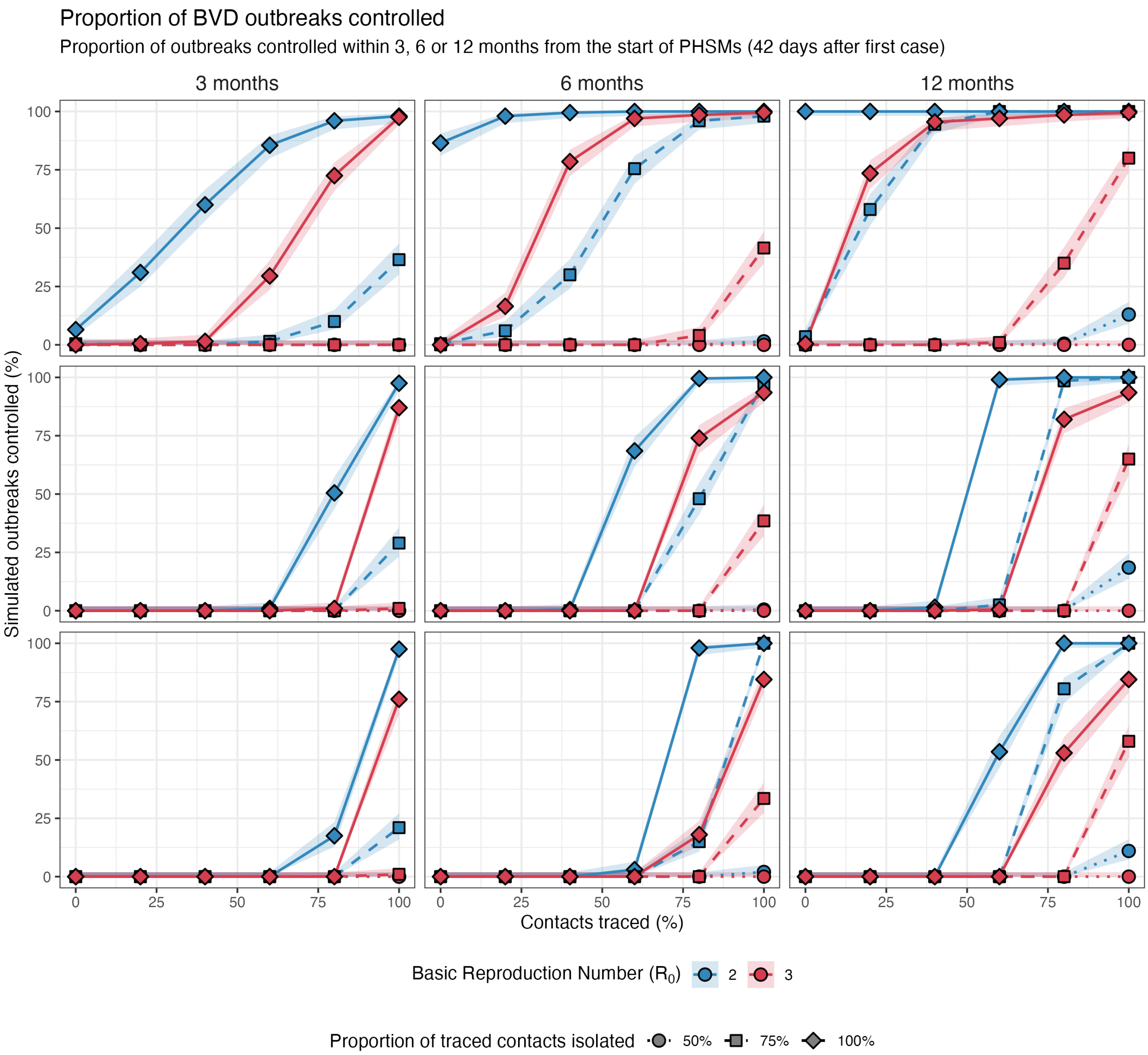

Figure 3: Proportion of simulated Bundibugyo virus outbreaks controlled (no new cases by the evaluation horizon) within 3, 6 and 12 months (columns) of PHSM activation, as a function of the percentage of infectious contacts traced. Rows show three mean onset-to-isolation delays (top to bottom: 1, 3 and 5 days; Gamma, SD = 1 day). Point shape and line style indicate three values of the proportion of untraced infections isolated: 50% (circle, dotted), 75% (square, dashed) and 100% (diamond, solid). Each point is the proportion of 200 conditioned simulations meeting the control criterion; shaded ribbons are 95% Wilson binomial confidence intervals around each proportion. PHSM activation is modelled as day 42 of the outbreak. Colours distinguish basic reproduction numbers R_0 = 2.0 and 3.0.

The descriptions of probability in this report correspond to those in the UK Government Probability Yardstick[16].

Discussion

Overview

In this report we have assessed the effectiveness of targeted PHSMs by identifying and isolating individuals with Ebola virus disease and tracing and isolating their contacts. We have evaluated how contact tracing coverage and speed of isolating symptomatic individuals impacts the probability of containment. By varying the speed of isolation we hope to cover responses that isolate suspected or probable cases, or those that require a positive test result[8]. We find that controlling the outbreak within 3 months is unlikely. Only with a fast (mean isolation <3 days from symptom onset to isolation) and comprehensive (>80% contacts traced) response do we find probable outbreak control within a year of interventions being active. Assuming community cases are always correctly identified shortens the time to elimination, with a realistic possibility of the outbreak lasting less than 6 months.

By incorporating the delay to activation of non-pharmaceutical measures we hope to more accurately estimate what control measures are effective and feasible for this outbreak, rather than assuming the intervention is activated by the index case, which would be the case if the outbreak were known and imported into a different region[11].

By evaluating outbreak control (extinction) at 3, 6 and 12 months from the activation of PHSMs, we provide a rough indication of the potential duration of the Bundibugyo outbreak. The majority of our simulated scenarios have a <50% chance of outbreak control within 6 months. Diagnostic sensitivity below approximately $1 - 1/R_0$ (≈50% at $R_0 = 2$, ≈67% at $R_0 = 3$) allows a self-sustaining chain of undiagnosed cases that contact tracing alone cannot interrupt, since traced contacts of an unisolated infector also remain outside the response in our model[17]. Together these results indicate that rapid development of pharmaceutical interventions, such as vaccines, may be required to help eliminate Bundibugyo transmission within the coming months[18].

Limitations

Our model does not account for the geographic spread of the Bundibugyo virus, and any geographical variation in response speed or capacity. Our analysis aims at informing what targeted PHSMs would be required to bring the outbreak to an end, and what outbreak duration is likely under different epidemiological characteristics until more precise estimates of parameters such as the basic reproduction number or the incubation period for Bundibugyo become available. There are reports of cases infected in healthcare facilities (∼50%) where contact tracing is less effective, and our model does not incorporate these setting-specific factors.

Given the limited estimation of epidemiological characteristics of the Bundibugyo virus in this outbreak we have based the incubation period and individual-level heterogeneity (k) on past estimates from the literature. Information on the transmission network would allow a better estimate of the individual-level heterogeneity in transmission[19]. If transmission is more homogeneous between people in the community then, theoretically, the outbreak would be less controllable[20]. We assess how sensitive our results are to changes in pathogen transmissibility (R_0), onset-to-isolation delays, and diagnostic sensitivity; however, these too are not precisely calibrated to early outbreak data[1], ongoing emergency response efforts, or reported test performance respectively. Additionally, in future work we could model the improved test sensitivity as new diagnostics specific to Bundibugyo become available.

We do not model capacity constraints on testing, isolating and contact tracing. These are resource-limited countermeasures, and if the number of suspected cases continues to rise these may become overwhelmed, reducing the efficacy of the intervention[17]. On 21 May 2026 the WHO reported contact tracing follow-up rate is 21%[21].

Our model assumes that traced contacts are followed up and correctly diagnosed. An alternative scenario would assume that diagnostic sensitivity applies equally to traced and untraced symptomatic infections; under that assumption the efficacy of contact tracing would be lower, and the structural sensitivity threshold described above would shift further from feasibility.

Conclusion

In conclusion, we hope that this report provides evidence to inform an effective control strategy for the ongoing epidemic caused by Bundibugyo ebolavirus using a mathematical modelling approach, as has been conducted for previous Ebola outbreaks[22]. Operationally, our findings suggest that the response should prioritise: (i) achieving high contact-tracing coverage as soon as practicable after activation; (ii) deploying diagnostic capacity with sensitivity adequate for Bundibugyo virus; and (iii) planning logistics for an outbreak that may persist for many months even under a well-resourced response.

References

[1] Institut National de Santé Publique de la République Démocratique du Congo. Situation report on Ebola virus disease (mve), no. 10, 2026. https://insp.cd/sitrep-mve-n-010-2026/, 2026. Accessed 2026-05-26.

[2] Centers for Disease Control and Prevention. Ebola situation summary. https://www.cdc.gov/ebola/situation-summary/index.html, 2026. Accessed 2026-05-27.

[3] World Health Organization. Epidemic of Ebola disease in the Democratic Republic of the Congo and Uganda determined a public health emergency of international concern. https://www.who.int/news/item/17-05-2026-epidemic-of-ebola-disease-in-the-democratic-republic-of-the-congo-and-ug May 2026. Accessed 2026-05-19.

[4] Ruth McCabe, L. Ebbarnezh, S. Okware, R. Fotsing, E. Koua, P. Mbaka, A. Lofungola, Sangeeta L. van Elsland, M. McMenamin, Neil M. Ferguson, O. le Polain de Waroux, and Anne Cori. Estimation of the size of the Ebola outbreak caused by Bundibugyo virus in the Democratic Republic of the Congo. Technical report, Imperial College London, May 2026. https://doi.org/10.25560/130053.

[5] Sam Abbott. BVDOutbreakSize. https://github.com/seabbs/BVDOutbreakSize, 2026. Accessed 2026-05-26. https://doi.org/10.5281/zenodo.20356667.

[6] Virological.org. Initial genomes from may 2026 bundibugyo virus disease outbreak in the democratic republic of the congo and uganda. https://virological.org/t/initial-genomes-from-may-2026-bundibugyo-virus-disease-outbreak-in-the-democratic-r 1032/5, 2026. Accessed 2026-05-25.

[7] Coalition for Epidemic Preparedness Innovations. Bundibugyo virus: What it is and what it is not. https://cepi.net/bundibugyo-virus-what-it-and-what-it-not, 2026. Accessed 2026-05-20.

[8] World Health Organization. Implementation and management of contact tracing for Ebola virus disease: emergency guideline. Emergency guideline WHO/EVD/Guidance/Contact/15.1, World Health Organization, Geneva, July 2015. URL https://iris.who.int/handle/10665/185258. Accessed 2026-05-25.

[9] Rebecca K. Nash, Sangeeta Bhatia, Christian Morgenstern, Patrick Doohan, David Jorgensen, Kelly McCain, Ruth McCabe, Dariya Nikitin, Alpha Forna, Gina Cuomo-Dannenburg, Joseph T. Hicks, Richard J. Sheppard, Thomas Naidoo, Sangeeta van Elsland, Cyril Geismar, Thomas Rawson, Sequoia I. Leuba, Jack Wardle, Isobel Routledge, and H. Juliette T. Unwin. Ebola virus disease mathematical models and epidemiological parameters: a systematic review. *The Lancet Infectious Diseases*, 24(12):e762–e773, 2024. doi: 10.1016/S1473-3099(24)00374-8.

[10] Jonathan A. Polonsky, Dankmar Böhning, Mory Keita, Steve Ahuka-Mundeke, Justus Nsio-Mbeta, Aaron Aruna Abedi, Mathias Mossoko, Janne Estill, Olivia Keiser, Laurent Kaiser, Zabulon Yoti, Patarawan Sangnawakij, Rattana Lerdsuwansri, and Victor J. del Rio Vilas. Novel use of capture–recapture methods to estimate completeness of contact tracing during an Ebola outbreak, Democratic Republic of the Congo, 2018–2020. *Emerging Infectious Diseases*, 27(12):3063–3072, 2021. doi: 10.3201/eid2712.204958.

[11] Joel Hellewell, Sam Abbott, Amy Gimma, Nikos I. Bosse, Christopher I. Jarvis, Timothy W. Russell, James D. Munday, Adam J. Kucharski, W. John Edmunds, Fiona Sun, Stefan Flasche, Billy J. Quilty, Nicholas Davies, Yang Liu, Samuel Clifford, Petra Klepac, Mark Jit, Charlie Diamond, Hamish Gibbs, Kevin van Zandvoort, Sebastian Funk, and Rosalind M. Eggo. Feasibility of controlling COVID-19 outbreaks by isolation of cases and contacts. *The Lancet Global Health*, 8(4):e488–e496, 2020. doi: 10.1016/S2214-109X(20)30074-7.

[12] WHO Ebola Response Team, Junerlyn Agua-Agum, Archchun Ariyarajah, Bruce Aylward, Isobel M. Blake, Richard Brennan, Anne Cori, Christl A. Donnelly, Ilaria Dorigatti, Christopher Dye, Tim Eckmanns, Neil M. Ferguson, Pierre Formenty, Christophe Fraser, Erika Garcia, Tini Garske, Wes Hinsley, David Holmes, Stephane Hugonnet, Srinivas Iyengar, Thibaut Jombart, Reshma Krishnan, Sara Meijers, Harriet L. Mills, Yasmine Mohamed, Gemma Nedjati-Gilani, Emily Newton, Pierre Nouvellet, Lyn Pelletier, Devin Perkins, Steven Riley, Maria Sagrado, Johannes Schnitzler, Dirk Schumacher, Anita Shah, Maria D. Van Kerkhove, Olivia Varsaneux, and Nimalan Wijesinghe Kannangarage. West African Ebola epidemic after one year — slowing but not yet under control. *The New England Journal of Medicine*, 372(6):584–587, February 2015. doi: 10.1056/NEJMc1414992.

[13] Centers for Disease Control and Prevention. About Ebola. `https://www.cdc.gov/ebola/about/index.html`, 2026. Accessed 2026-05-26.

[14] Placide Mbala, Marc Baguelin, Ipos Ngay, Alicia Rosello, Prime Mulembakani, Nikolaos Demiris, W. John Edmunds, and Jean-Jacques Muyembe. Evaluating the frequency of asymptomatic Ebola virus infection. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 372(1721): 20160303, May 2017. doi: 10.1098/rstb.2016.0303.

[15] M. Jana Broadhurst, Tim J. G. Brooks, and Nira R. Pollock. Diagnosis of Ebola virus disease: Past, present, and future. *Clinical Microbiology Reviews*, 29(4):773–793, October 2016. doi: 10.1128/CMR.00103-15.

[16] Professional Head of Intelligence Analysis. Explaining uncertainty in UK intelligence assessment. `https://www.gov.uk/government/publications/explaining-uncertainty-in-uk-intelligence-assessment/explaining-uncertainty-in-uk-intelligence-assessment`, March 2025. Accessed 2026-05-26.

[17] Ranu S. Dhillon and Devabhaktuni Srikrishna. When is contact tracing not enough to stop an outbreak? *The Lancet Infectious Diseases*, 18(12):1302–1304, December 2018. doi: 10.1016/S1473-3099(18)30656-X. `https://doi.org/10.1016/S1473-3099(18)30656-X`.

[18] Oxford Vaccine Group, University of Oxford. Statement on vaccine efforts relating to the Bundibugyo ebolavirus outbreak in the DRC. https://www.ox.ac.uk/news/2026-05-22-statement-on-vaccine-efforts-relating-to-the-bundibugyo-ebolavirus-outbr May 2026. Accessed 2026-05-27.

[19] Christian L Althaus. Ebola superspreading. *The Lancet Infectious Diseases*, 15(5):507–508, May 2015. ISSN 1473-3099. doi: 10.1016/s1473-3099(15)70135-0. URL http://dx.doi.org/10.1016/S1473-3099(15)70135-0.

[20] J. O. Lloyd-Smith, S. J. Schreiber, P. E. Kopp, and W. M. Getz. Superspreading and the effect of individual variation on disease emergence. *Nature*, 438(7066):355–359, November 2005. doi: 10.1038/nature04153.

[21] World Health Organization. Ebola disease caused by Bundibugyo virus – Democratic Republic of the Congo. Disease Outbreak News. https://www.who.int/emergencies/disease-outbreak-news/item/2026-DON603, May 2026. Accessed 2026-05-27.

[22] Adam J. Kucharski, Rosalind M. Eggo, Conall H. Watson, Anton Camacho, Sebastian Funk, and W. John Edmunds. Effectiveness of ring vaccination as control strategy for Ebola virus disease. *Emerging Infectious Diseases*, 22(1):105–108, January 2016. doi: 10.3201/eid2201.151410. https://doi.org/10.3201/eid2201.151410.

Code availability

The code to reproduce this analysis can be found at: https://github.com/joshwlambert/BundibugyoNPI2026

The outbreak model code can be found in the R package `ringbp` at https://github.com/joshwlambert/ringbp@develop. This analysis used the `develop` branch of the ringbp package which can be installed with:

```
remotes::install_github("joshwlambert/ringbp@develop")
```

AI Disclaimer

This work was assisted by the use of Claude Opus 4.7.

Supplementary Material

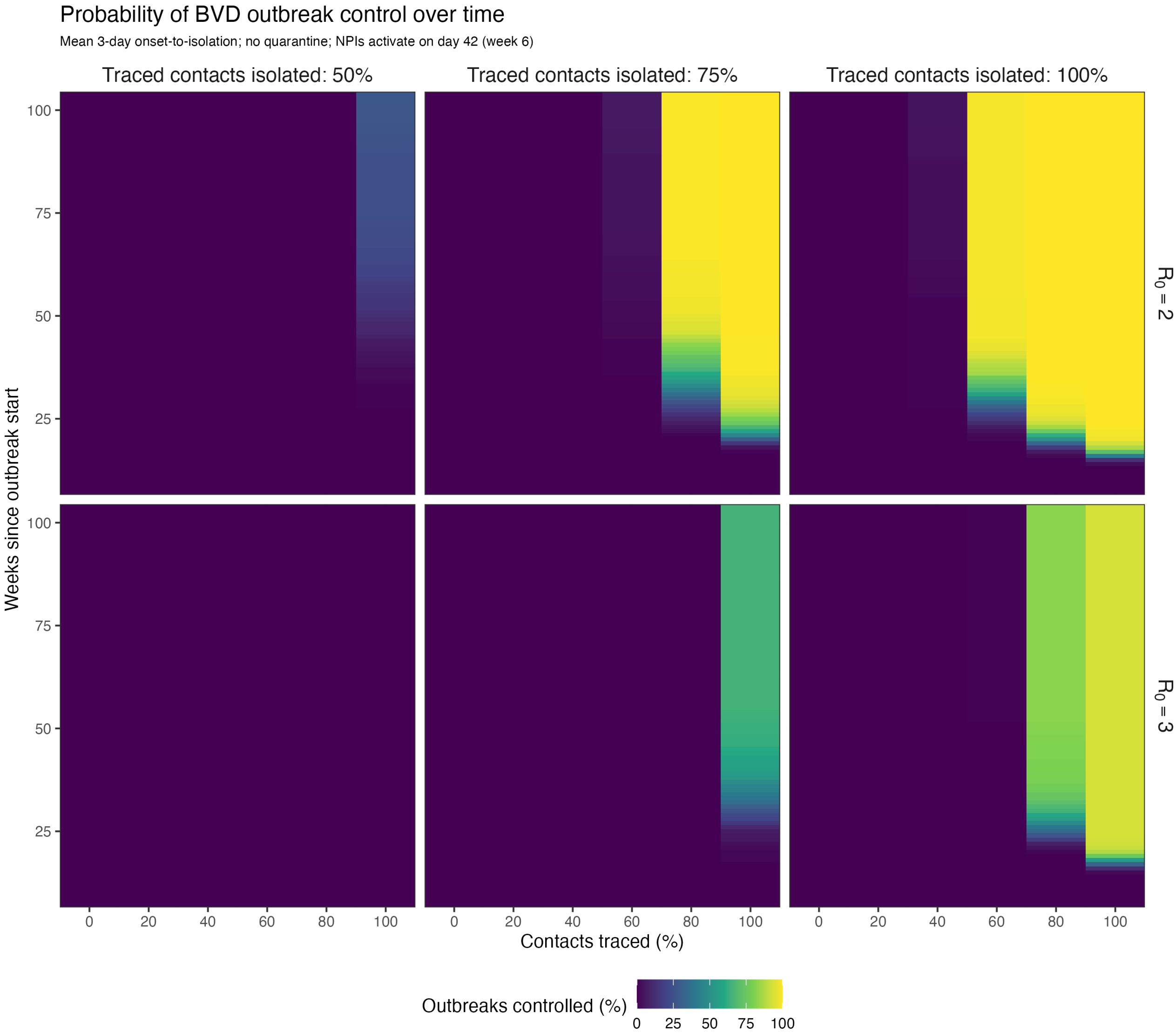

Figure 4: Probability of Bundibugyo virus disease outbreak control as a function of the percentage of contacts traced (x-axis) and the time elapsed since the outbreak start (y-axis, weeks). Rows correspond to the assumed basic reproduction number ($R_0 = 2$ and $R_0 = 3$); columns correspond to the assumed test sensitivity (50%, 75% and 100%). The fill colour shows the proportion of conditioned simulations that had reached extinction by the given week. Restricted to the mean 3-day onset-to-isolation scenario. PHSMs activate on day 42 of the outbreak (week 6).